

Kenechukwu Emmanuel Nduaguba

FULL-STACK DEVELOPER & AI / AUTOMATION ENGINEER

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Full-stack developer and automation engineer with 4+ years shipping production software across fintech, telecom, and SaaS for African markets. Specializes in AI-powered workflow automation (n8n, LangChain, Google ADK), payment infrastructure (Paystack), and scalable API design (FastAPI, Next.js) — pairing backend fundamentals with a product-first mindset and deep fluency in the Nigerian tech ecosystem.

EXPERIENCE

Software Engineer — Automation & Integrations

2024 — Present

Tecnotree Corporation

- Designed and deployed n8n automation workflows for MTN Uganda's telecom billing operations, surfacing real-time ServiceNow incident data into Microsoft Teams and cutting manual triage turnaround.
- Built a multi-step incident-management reporting system on ServiceNow REST APIs, replacing manual reporting and reducing reporting lag.
- Evaluated and scoped lien-management system architecture for TSAT, DCLM, and MA/DA wallet scenarios.
- Collaborated with engineering teams across Africa and Europe on CMS improvements.

Senior TechOps Engineer — Application Support

2022 — 2024

Tecnotree, Lagos

- Ran root-cause analysis on critical production incidents across multiple live telecom systems, keeping service disruption minimal.
- Led AI-driven automation initiatives that improved operational efficiency and reduced manual workload.
- Managed MTN Nigeria user-access audits for system security, compliance, and data integrity.
- Partnered with MTN Nigeria Opco to optimize BSS/OSS workflows for faster service delivery.

Application Support Intern

Oct 2021 — 2022

Tecnotree, Lagos

- Supported testing and validation of Tecnotree Moments products; documented the Auto Direct Debit solution to speed onboarding.
- Assisted in application monitoring and timely issue resolution.

SELECTED PROJECTS

LinkedIn Job Finder — Job Search SaaS

[linkedin-job-finder-phi.vercel.app](#)

React · Supabase (Postgres, RLS) · n8n · Apify · Vercel · Paystack

- Designed and shipped a multi-tenant SaaS end to end — email/password auth with row-level security, per-plan usage quotas enforced server-side, a Kanban application tracker, and CSV export — from empty repo to production.
- Architected an asynchronous search pipeline (webhook → background worker → polled run status) so a two-minute scrape never holds a browser connection open, with a shared result cache that serves repeat queries instantly and cuts scraping spend per user.
- Engineered a scheduled alert engine that diffs every run against per-user delivery history, so subscribers only ever receive postings they have not been sent, alongside follow-up reminders and weekly pipeline digests.
- Instrumented for operations: Telegram failure alerts, daily health checks that surface silent failures, automated backups, Sentry error capture, and privacy-first product analytics.

Automated Client Onboarding Workflow

n8n · Form Trigger · Google Drive API · ClickUp API · Gmail · Slack · Google Sheets

- Engineered an end-to-end onboarding automation triggered by an n8n intake form, provisioning a structured Google Drive workspace (Contracts, Assets, Deliverables, Invoices) and a ClickUp project pre-loaded with standardized starter tasks for each new client.
- Reduced manual onboarding administration from 30–60 minutes to under 30 seconds per client, synchronizing the internal team via automated Slack notifications and a Google Sheets audit log.
- Resolved response-handling constraints specific to n8n's Form Trigger architecture (completion-node vs. webhook responses) to deliver distinct success and error states for the client-facing form.

AI-Powered Architecture Firm Website & Knowledge-Base Chatbot

n8n · Pinecone · OpenAI Embeddings · RAG · Docx automation

- Delivered a client-facing site plus a RAG chatbot answering visitor questions on the firm, its services, and the end-to-end house-building process.

- Authored a structured company knowledge base and indexed it into Pinecone through an n8n ingestion pipeline for grounded retrieval.
- Diagnosed an embedding/index vector-dimension mismatch and tuned the agent prompt to scope in-domain vs. out-of-domain queries without over-refusing.

AI-Powered LinkedIn Content Automation Pipeline

n8n · LinkedIn API (OAuth2) · LLM agent · AI image generation · Google Sheets

- Built an end-to-end pipeline that researches trending topics, synthesizes an original on-brand post, generates a matching illustration, and auto-publishes via the LinkedIn API.
- Logged full post history to Google Sheets for tracking and analytics.
- Solved OAuth2 scope mismatches, binary image handling, and enforced structured JSON output against inconsistent API responses.

DieselGuard — ML Diesel-Loss Detection SaaS

Python · Isolation Forest · FastAPI · Next.js · Supabase · Recharts · Telegram

- Built an anomaly-detection system using Isolation Forest to flag diesel theft and loss in generator operations.
- Shipped end to end: FastAPI backend, Next.js dashboard with Recharts, Supabase database, Telegram alerts, and PDF export.

Job Application Triage Agent

Google ADK 2.0 · LiteLlm / GPT-4o-mini · Python — Kaggle capstone, July 2026

- Built an autonomous agent on Google's Agent Development Kit that classifies, prioritizes, and routes incoming job applications through an LLM reasoning pipeline; capstone for the Google DeepMind ADK curriculum.

AI Web Scraper & Netflix Movie Finder

Python · BeautifulSoup · Groq (Llama 3.3 70B) · LangChain · ChromaDB · Streamlit

- webscrappai.streamlit.app — scrape any page and chat with its content; auto-summarization, metadata cards, history, Excel/CSV export, token tracking.
- netflixfinder.streamlit.app — RAG semantic search over 8,800+ Netflix titles by vibe or story description, embedded with HuggingFace BGE in ChromaDB.

InvoiceNaija — SaaS Invoicing Platform

Next.js · Supabase · Paystack · Railway

- Nigerian invoicing SaaS with WhatsApp PDF sharing, Paystack subscription billing, and multi-currency invoices — zero to production in a single build session.

TECHNICAL SKILLS

Languages	Python, JavaScript (ES6+), TypeScript, SQL, HTML/CSS
Frameworks	Next.js, React, FastAPI, Node.js, Vite
AI / ML	LLMs, agent orchestration, LangChain, Google ADK, RAG systems, n8n AI agents, HuggingFace, ChromaDB, Groq, OpenAI GPT-4o, Playwright
Automation	n8n (advanced), webhook orchestration, async job pipelines, API integration
Data	PostgreSQL, Supabase (RLS, RPC, views), pgvector, SQLite, schema design, query optimization
Payments	Paystack (virtual accounts, subscriptions), Stripe webhooks, Reloadly
Messaging	WhatsApp Cloud API, Telegram Bot API, Microsoft Teams webhooks, LinkedIn API, SMTP/Resend
CRM	HubSpot & HubSpot API, Salesforce API, ServiceNow REST API
DevOps	Docker, Kubernetes, AWS EC2, GitHub Actions, Vercel, Render, Railway, Streamlit Cloud, REST API design
Practices	Multi-tenant architecture, row-level security, GDPR/NDPA compliance, error monitoring, product analytics

EDUCATION & CERTIFICATIONS

Higher National Diploma, Computer Science

Federal Polytechnic Oko, Anambra State · 2018

National Diploma, Computer Science

Federal Polytechnic Oko, Anambra State · 2015

Founder, KCEMMA HUB

(RC 9602233) — registered Nigerian business offering AI automation services, WhatsApp commerce tooling, and technical training.

- n8n Foundations: AI, Testing & Best Practices — n8n, 2026
- n8n Foundations: Integrations & APIs — n8n, 2026
- TRI AI Saturdays Cohort 10, Google DeepMind curriculum — in progress
- AWS AI & ML Scholars — 2026 challenge completion
- Google IT Support Professional Certificate
- HubSpot Academy — Digital Marketing Certified
- Scrum Fundamentals · Aviatrix Certified Engineer · Python for Everybody Specialization